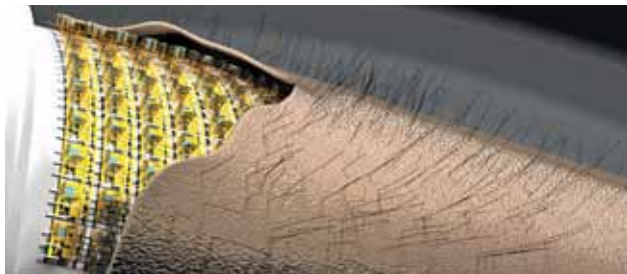




location, two cameras, lights, and a manipulator arm for swabbing radioactive pollution from the duct's wall or floor. Lyra's radiation sensor package was built to measure beta, gamma, x-ray, and neutron radiations. Its front and the manipulator's end were both fitted with cameras. Lyra is controlled via a joypad for driving and a supply manipulator arm whose motion is replicated by the robot's arm. This allowed for the creation of a 3D, time stamped video with the measured radiation readings overlayed onto the video.

Researchers develop super-sensitive e-skin

A research team from Chemnitz and Dresden has made significant progress in the development of sensitive



Artificial electronic skin (Credit: <https://www.tu-chemnitz.de/tu/>)

electronic skin (e-skin) with integrated artificial hair. The team looked at a new way to develop very sensitive and direction-dependent 3D magnetic field sensors that may be used in an e-skin system. An anisotropic magnetoresistance (AMR) sensor lies at the heart of the sensor system, which was created using the 'micro-origami method.' Similar to real skin, which is interwoven with nerves, this e-skin is made of an elastomeric material into which the electronics and sensors are placed.

AI-enabled bot that cleans water bodies

Clearbot Neo is a sleek AI-enabled robotic boat designed by Open Ocean Engineering, a Hong Kong based startup, that gathers tonnes of floating debris. The bot, which is only three metres long and is propelled by a solar battery-

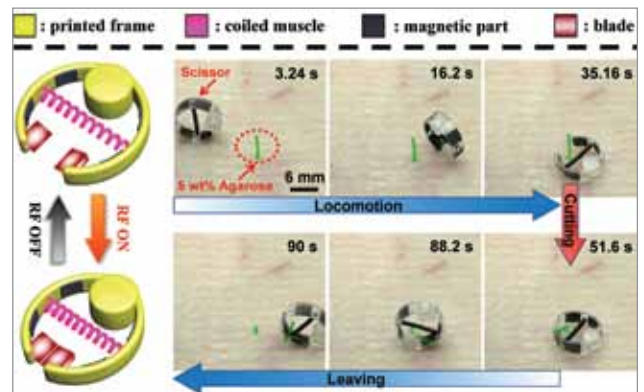


AI-enabled robotic boat by Open Ocean Engineering, Clearbot Neo, collects data to track and tackle pollution at source using Microsoft Azure (Credit: Clearbot)

powered electric motor, systematically skims the surface and sweeps up floating debris. It uses artificial intelligence to distinguish and log the types of rubbish it collects and where it finds it. It can collect data for companies, governments, non-governmental organisations, and individuals using Microsoft Azure and monitor the origins of marine debris. Subsequently, it can combat pollution at the source.

Soft bots to perform surgery

Researchers at Stuttgart's Max Planck Institute for Intelligent Systems have developed a small-scale coiled muscle actuator that uses tunable mechanical properties and radio frequency magnetic heating to perform surgical tasks that require a high force output and work capacity, such as suturing,



Magnetic control: A scissoring device rolls to the target location and cuts a tissue-like structure using radiofrequency heating (Credit: <https://physicsworld.com/>)

cutting, drilling, and clamping tissues. "Our new approach extends the capabilities of wireless medical soft robots to surgical applications. As next step, we will demonstrate such functions in small animal surgeries in *in-vivo* conditions," explain senior authors Wenqi Hu and Metin Sitti.

Lunar colony being planned for Mars exploration

NASA's planned Artemis lunar colony will serve as a ground for eventual human exploration of Mars. The base camp will have independent mining and fuel processing facilities. Sandia National Laboratories is creating an electrical system controller for the mining and processing centre's microgrid. According to Jack Flicker, a Sandia electrical engineer, NASA is designing the electrical system controller for the habitation unit because the system will be quite similar to the direct-current electrical system on the International Space Station. Flicker and his colleagues are working on the technology that will connect the two microgrids as well as the power flow and operation between them.